Spheroid Metrics

Automated Segmentation and Morphometric Analysis of Bright-Field Images Cancer Spheroids

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Key Stakeholder

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Team

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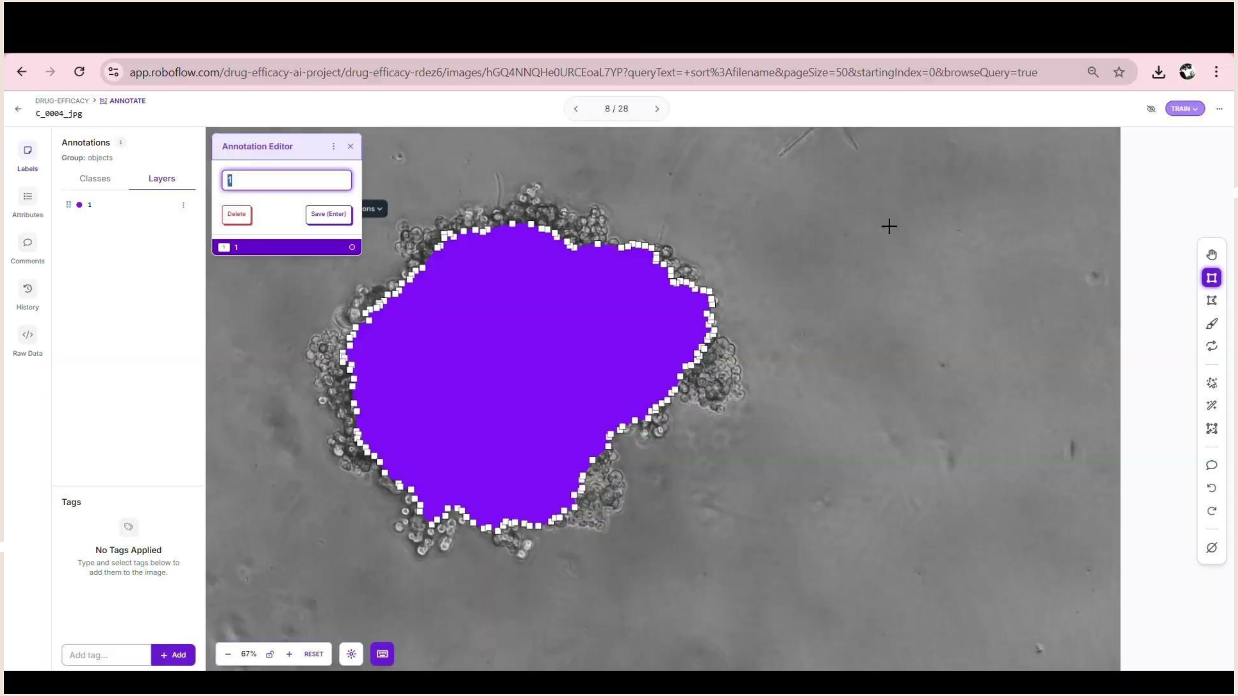
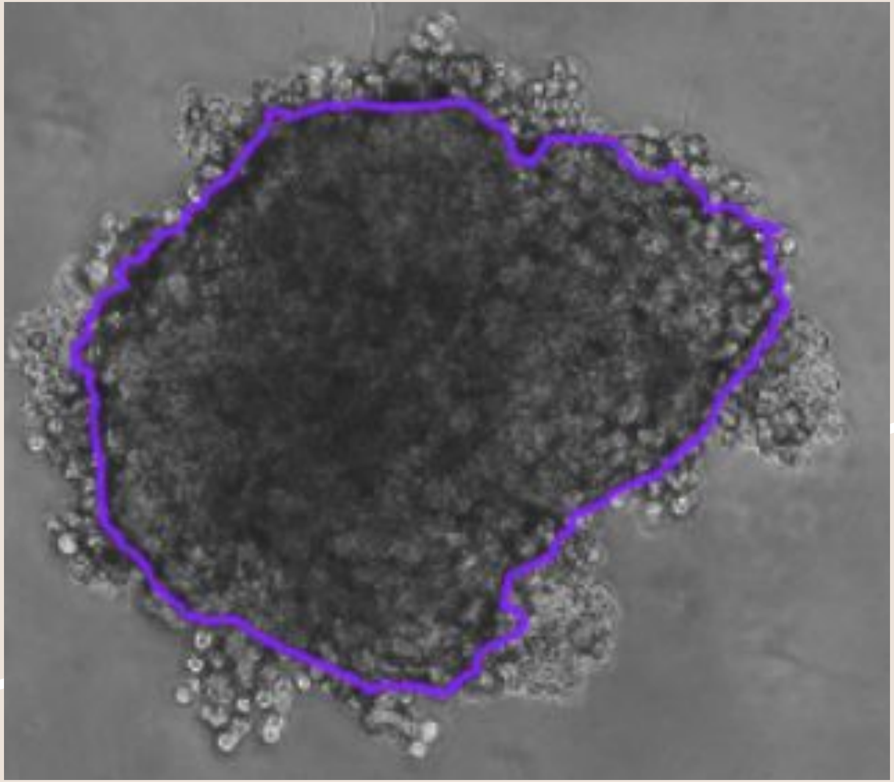
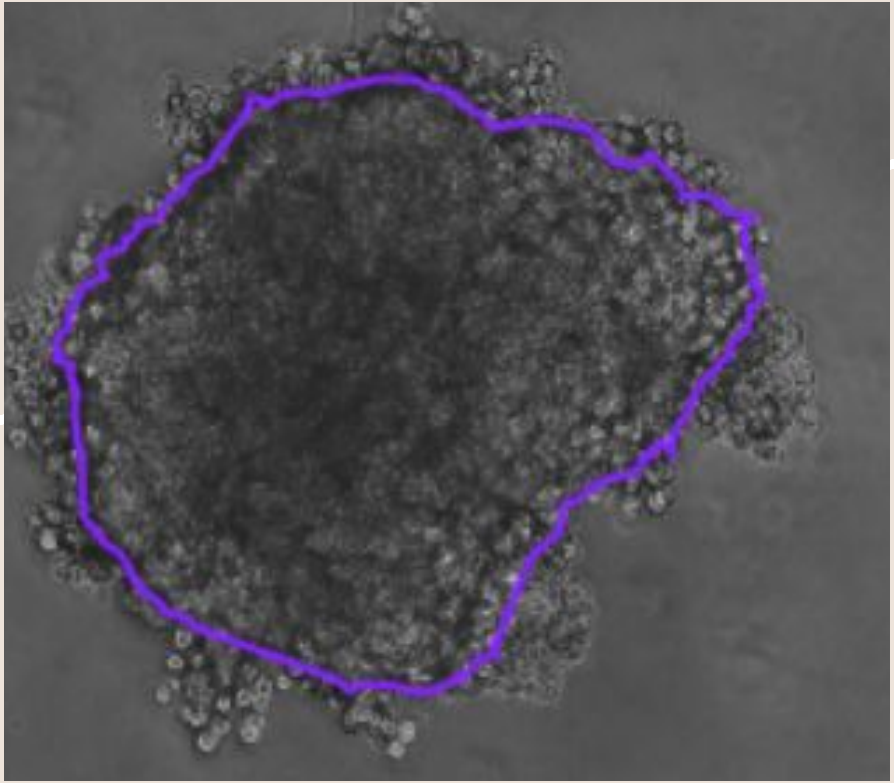
Manual annotation

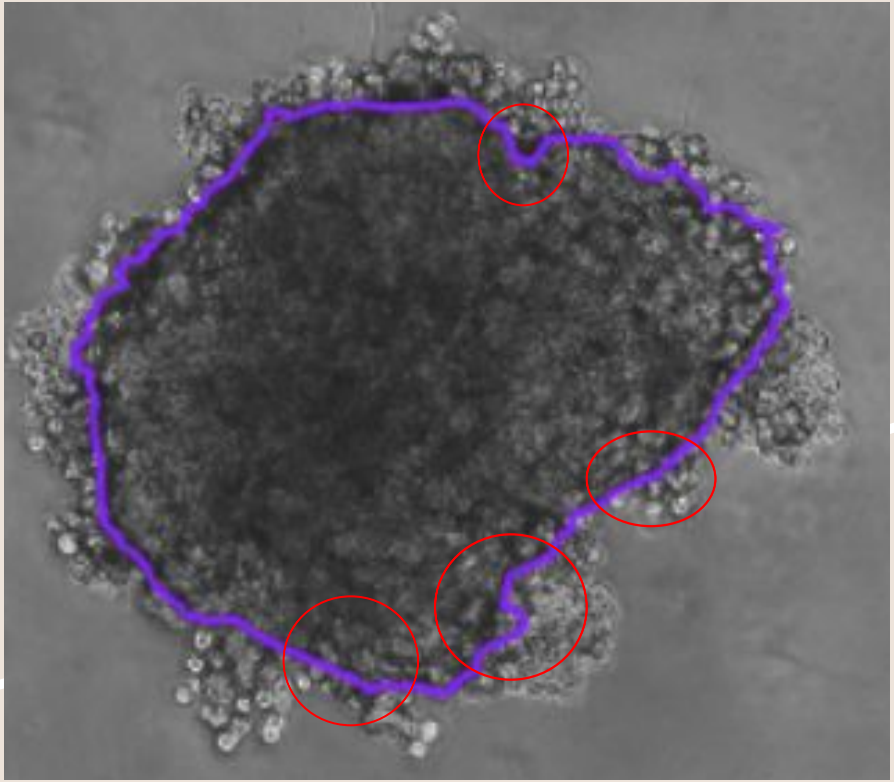


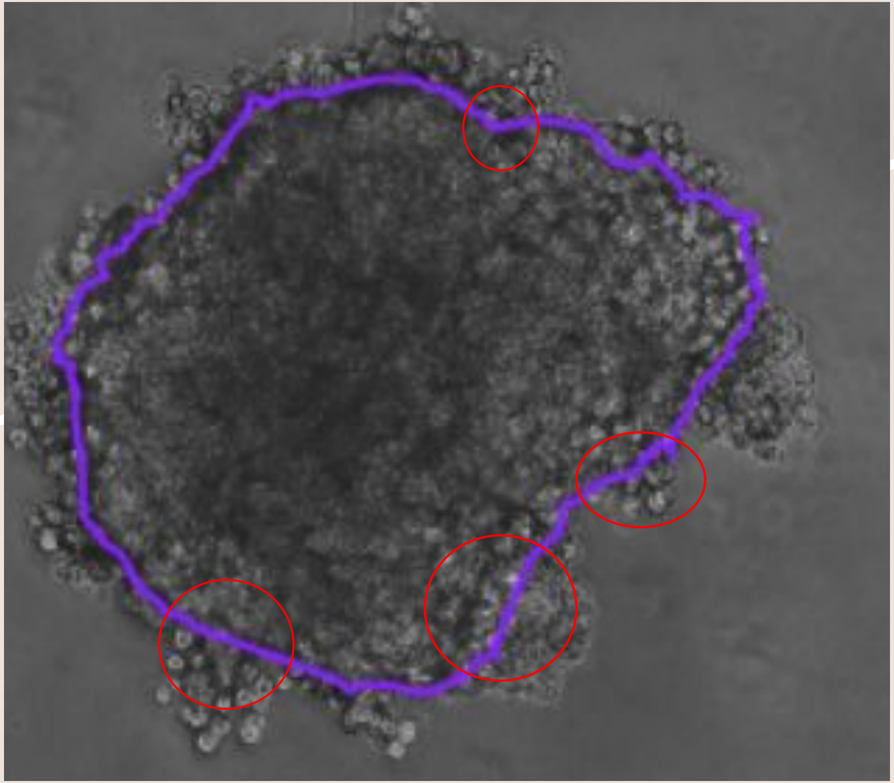
Spot the differences





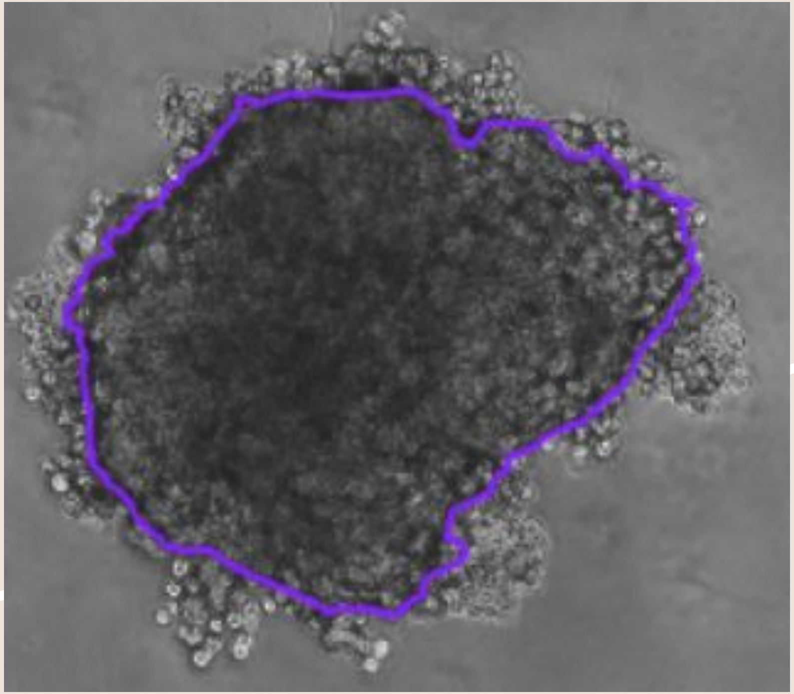
Spot the differences



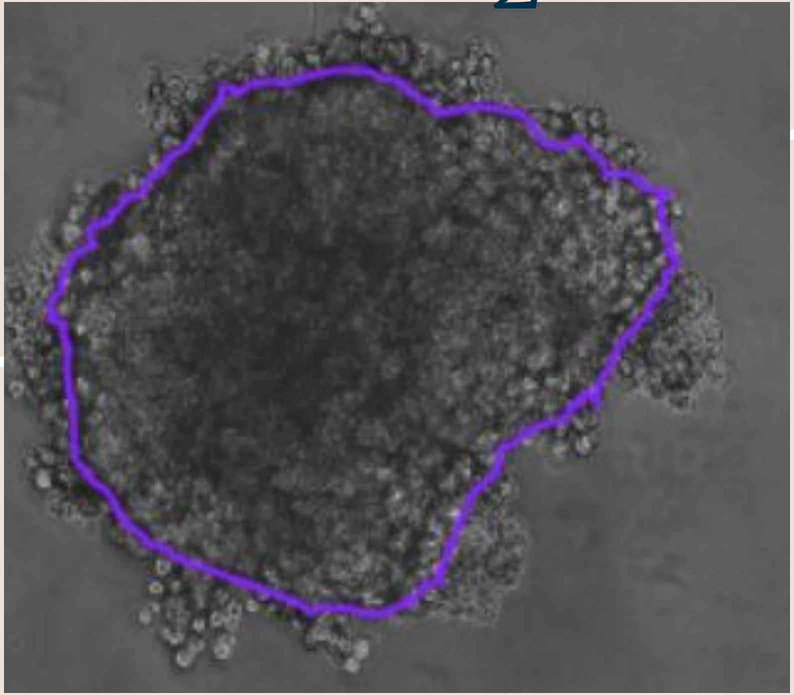


Select the correct annotation

1



2



Problem

1. **Pancreatic cancer**: fatal, hard to treat.
2. 3D spheroids = realistic models (mimic the microenvironments and cell-cell interactions of actual cancer).
3. Manual annotation and analysis is slow and subjective - Not Ideal for scalability.
4. AI + drug Analysis= Fast and Efficient.

Goal: AI Tool for drug analysis on bright field images of 3D spheroids

Objective

1 Image Segmentation

Implement and optimize a segmentation method to accurately detect spheroid boundaries from bright-field microscope images.

2 Feature Extraction

Calculate key morphological/Intensity metrics such as,

- Area
- Perimeter
- Circularity
- GLCM features

from the segmented part to quantify growth/treatment-induced changes in spheroid structure

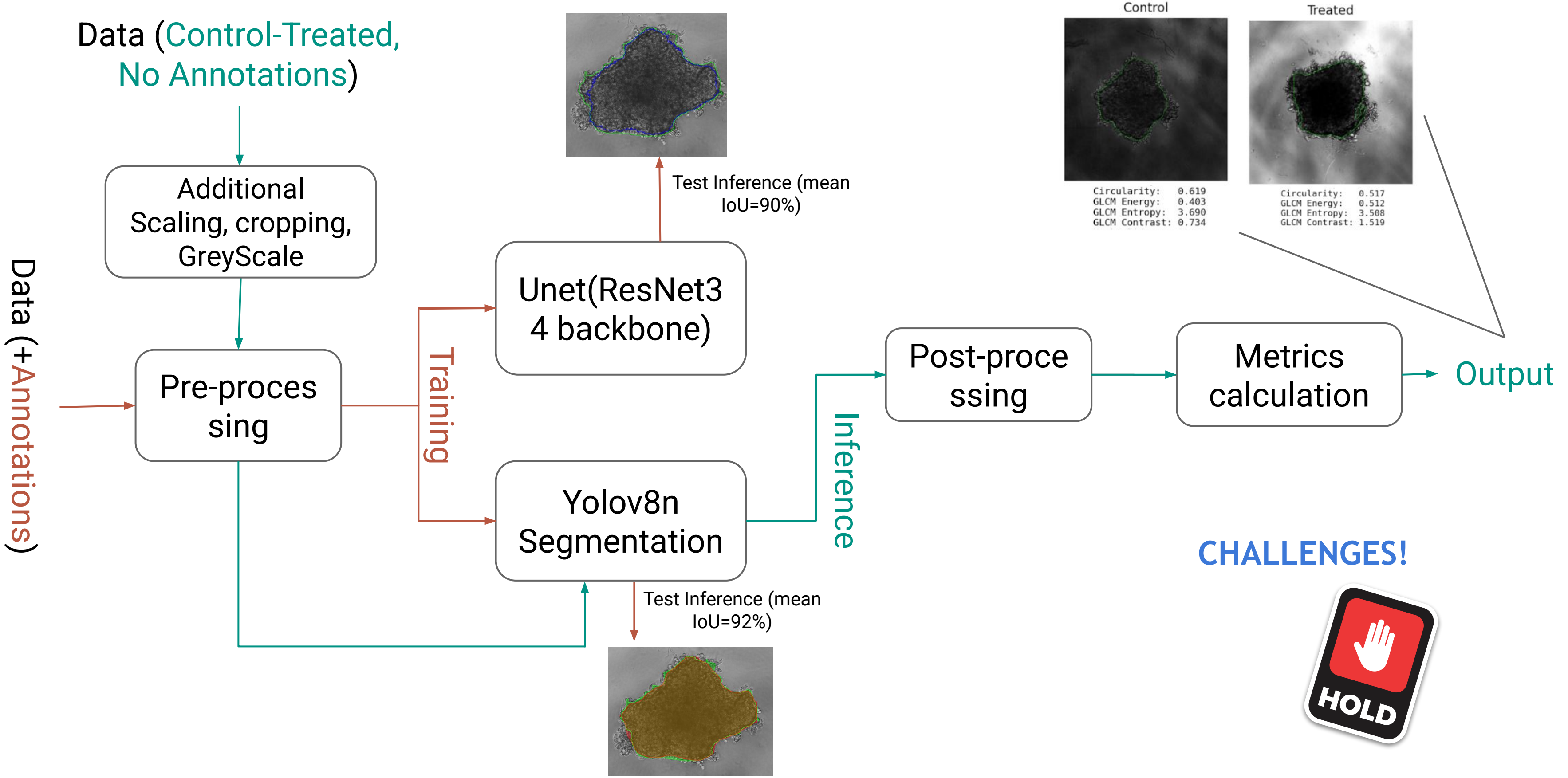
3 Feature Comparison

Perform comparative analysis of extracted metrics between images taken before and after drug treatment.

Data

- Bright field Images of in vitro PDAC spheroids (pancreatic ductal adenocarcinoma).
- Annotated Data -179.
- Unannotated Data (Control-Treated pair)- 60.
- Control(Untreated) images captured on day 7.
- Treated(Treated with drug combination JQ1 +GANT61) images captured on day 10.

Pipeline


Data (Control-Treated, No Annotations)
Additional Scaling, cropping, GreyScale
Data (+Annotations)
Pre-processing
Training
Unet(ResNet34 backbone)
Test Inference (mean IoU=90%)
Yolov8n Segmentation
Test Inference (mean IoU=92%)
Inference
Post-processing
Metrics calculation
Output
Control
Circularity: 0.619
GLCM Energy: 0.403
GLCM Entropy: 3.690
GLCM Contrast: 0.734
Treated
Circularity: 0.517
GLCM Energy: 0.512
GLCM Entropy: 3.508
GLCM Contrast: 1.519
CHALLENGES!
HOLD

Segmentation

Attribute	U-Net	YOLOv8n
Base Architecture	Encoder–Decoder with skip connections	Unified detection–segmentation network
Encoder Backbone	ResNet-34 (pretrained)	YOLOv8n-seg (native)
Segmentation Type	Pixel-wise (semantic)	Instance-level
Input Resolution	512 × 512	768 × 768
Epochs	30	1,000
Optimizer	Adam	Ultralytics default pipeline
Learning Rate	1×10^{-4}	Not specified

Segmentation

Attribute	U-Net	YOLOv8n
Loss Formula	L = α_BCE·L_BCE + α_IoU·L_IoU + α_Boundary·L_Boundary + α_Area·L_Area	L = L_box + L_cls + L_mask + L_dfl
Mean IoU	0.90	0.92
Area Ratio (GT vs Pred)	~98.5%	~99%
Boundary Fidelity	Smoother in high-contrast; reduced boundary IoU under low contrast	Robust in cluttered/low-contrast conditions
Robustness	Sensitive to boundary ambiguity & low contrast	Robust to debris, merged objects, heterogeneous backgrounds
Selected for Inference?	❌ No	✅ Yes

Metrics ??

That helps track the changes and are biologically interpretable??

Morphological Metrics

1. **Area** — $A = \Sigma M(x,y)$ — Decrease indicates growth inhibition or structural disintegration.
2. **Perimeter** — Boundary length — Disproportionate increase relative to area indicates reduced compactness.
3. **Circularity** — $C = 4\pi \cdot \text{Area}/\text{Perimeter}^2$ — Lower values indicate morphological instability and structural degradation.

Intensity Metrics (GLCM Texture Features)

1. **Energy** — $E = \Sigma P(i,j)^2$ — Drops when internal texture becomes less uniform after treatment.
2. **Entropy** — $H = -\Sigma P(i,j) \cdot \log_2(P(i,j))$ — Rises with increased randomness and disorder inside the spheroid.
3. **Contrast** — $C = \Sigma (i-j)^2 \cdot P(i,j)$ — Higher values reflect greater local intensity variation within the spheroid.

GLCM (Gray-Level Co-Occurrence Matrix)

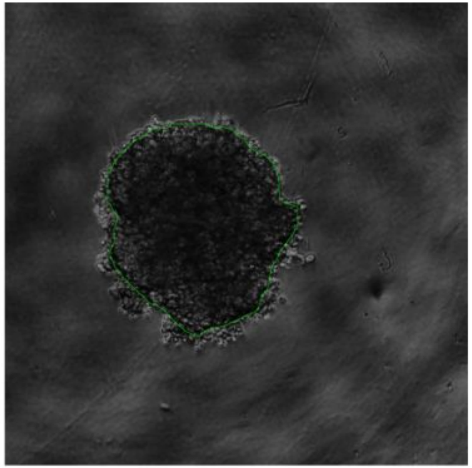
1. Create a matrix containing the grayscale pixel values of the relevant region of the image (background excluded)
2. Quantize pixel values into 32 levels
   a. 255 divided by 32 -> pixel values replaced by quantized levels
      i. 0 - 7 -> level 0
      ii. 8 - 15 -> level 1
      iii. ...
      iv. 248 - 255 -> level 31
3. Choose distance and angle
   a. Ex. distance = 1, angle = 0
4. Count co-occurrences per row
5. Build the GLCM matrix where each entry maps how many times a level appears next to another level, this matrix will be a 32 x 32 matrix
6. Normalize the GLCM matrix by the total number of pairs to get the probabilities of each pairing

$$\sum_{i,j} P(i,j) = 1$$

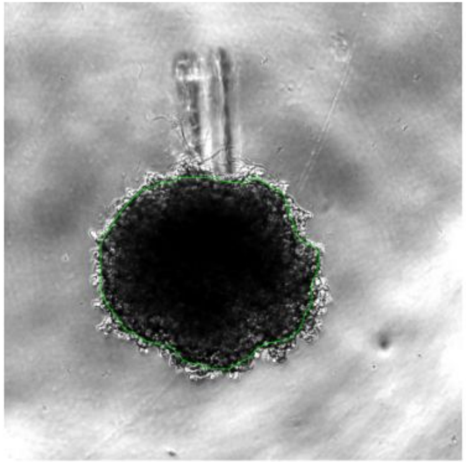
Visual Results: Control v. Treated

G_0010

Control



Treated

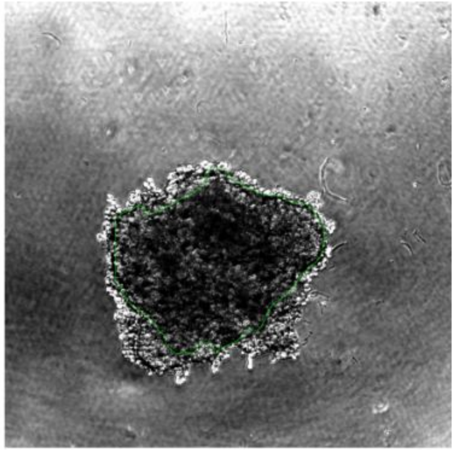


Control		Treated	
Circularity:	0.641	Circularity:	0.717
GLCM Energy:	0.360	GLCM Energy:	0.509
GLCM Entropy:	4.093	GLCM Entropy:	3.995
GLCM Contrast:	1.307	GLCM Contrast:	4.921

Visual Results: Control v. Treated

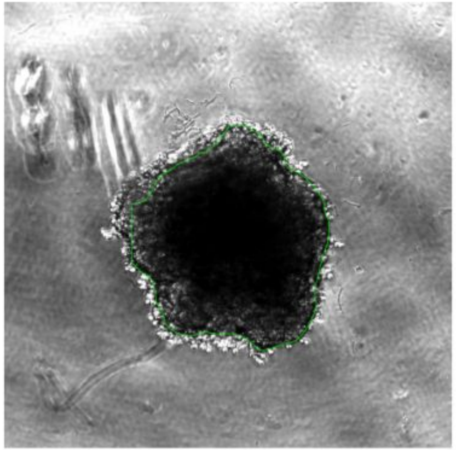
F_0011

Control



Circularity: 0.541
GLCM Energy: 0.479
GLCM Entropy: 4.639
GLCM Contrast: 6.369

Treated



Circularity: 0.681
GLCM Energy: 0.522
GLCM Entropy: 3.944
GLCM Contrast: 3.547

Tabular Results: Control v. Treated

Metric	F_0011 (% change)	G_0010 (% change)
Circularity	+25.9%	+11.9%
GLCM Energy	+9.0%	+41.4%
GLCM Entropy	-15.0%	-2.4%
GLCM Contrast	-44.3%	+276.6%

Limitations

- The selected metrics reflect shape and texture variations but do not capture complex biological processes such as viability, invasion, or multilayer structure.

- The framework serves as an interpretable baseline for initial screening of bright-field spheroid assays, not as a full mechanistic analysis solution.

- The current implementation is designed for technically skilled users capable of directly running and managing the analysis pipeline.

Thank you